

PolyAgora Phase-II Specification

Recoverable vs. Irreversible Convexity Architecture

MRTP Convexity Governance Layer

0. Executive Thesis

PolyAgora's current architecture already demonstrates a strong capability in:

- survivability,
- drawdown compression,
- admissibility governance,
- recoverability preservation under rupture.

However, the same governance machinery structurally suppresses participation during persistent convex expansion regimes.

This is not an implementation bug.

It is the direct consequence of a deeper architectural fact:

Current PolyAgora treats convexity itself as potentially recoverability-destructive.

Phase II therefore introduces a critical distinction:

Two Classes of Convexity

1. Irreversible Convexity
(path-fragile, leverage-fragile, forced-traversal dependent)
2. Recoverable Convexity
(optionality-preserving, bounded-risk, corridor-compatible)

The purpose of the new layer is:

maximize admissible convexity subject to future maneuverability preservation.

This becomes the governing MRTP principle for expansion regimes.

1. The Core Challenge

Why V7/V10 outperform in crises

Current PolyAgora correctly prioritizes:

- admissibility,
- recoverability,
- corridor preservation,
- anti-fragility,
- controlled contraction.

This creates:

- strong GFC behavior,
- strong COVID behavior,
- strong 2022 behavior,
- low drawdown,
- low SPY correlation.

But it also creates:

- delayed re-risking,
- cautious re-entry,
- suppressed beta participation,
- reduced convex expansion capture.

The system currently optimizes:

Recoverability » Convex Exploitation

Phase II changes this relationship.

2. The Key Insight

Not all convexity destroys recoverability.

Current architecture implicitly assumes:

Convexity → Higher Recoverability Risk

But this is only true for certain exposure structures.

The correct decomposition is:

Convexity = Recoverable Convexity + Irreversible Convexity

This distinction becomes a first-class runtime governance object.

3. Irreversible Convexity

Definition

Exposure structures that:

- require future forced traversal,
 - depend on liquidity continuity,
 - create asymmetric collapse risk,
 - narrow future admissible corridors,
 - reduce future maneuverability,
 - increase ruin sensitivity.
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Examples

A. Excessive directional leverage

- high beta expansion,
- margin-sensitive exposure,
- path dependency.

B. Naked short exposure

- future mandatory buyback,
- squeeze sensitivity,
- forced future liquidity requirement.

C. Concentrated path-dependent trades

- narrow admissibility corridor,
- inability to rotate.

D. Correlation-collapse exposure

- assumes stable cross-asset structure,
 - fails under regime transition.
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MRTP Interpretation

These structures violate:

Future Reachability Preservation

They optimize local payoff while damaging:

- re-entry capacity,
- future steering,

- corridor width,
 - recoverability geometry.
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4. Recoverable Convexity

Definition

Exposure structures that:

- preserve future maneuverability,
 - maintain optionality,
 - bound downside,
 - widen future admissible corridors,
 - permit controlled contraction,
 - preserve steering flexibility.
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Examples

A. Long optionality structures

- bounded downside,
- convex upside,
- no forced traversal.

B. Cross-sectional dispersion

- long winners / short losers,
- low directional beta,
- exploit expansion geometry without full market dependency.

C. Transition breakout sleeves

- early expansion participation,
- only under low fragmentation.

D. Recoverability-aware momentum

- momentum gated by corridor width,
- disabled during fragmentation spikes.

E. Crisis-to-reflation asymmetry sleeves

- controlled convex participation post-rupture.

M RTP Interpretation

These structures preserve:

Admissible Future Reachability

while still permitting:

- expansion,
- convex payoff,
- corridor participation.

5. New Layer: Exposure Recoverability Quality (ERQ)

New Runtime Layer

Introduce:

Layer 5A — Exposure Recoverability Geometry

This layer evaluates not:

- expected return,

but:

- future maneuverability preservation quality.

6. ERQ Score

Each exposure receives:

$ERQ_i \in [0, 1]$

Where:

Score	Meaning
0	recoverability-destructive
1	recoverability-preserving

7. Proposed ERQ Components

A. Forced Traversal Risk

Question:

does this exposure require future market cooperation to exit safely?

Examples:

- short squeeze risk,
 - leverage unwind risk.
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B. Corridor Width Preservation

Question:

does this exposure preserve multiple future admissible trajectories?

High ERQ:

- optional structures,
- bounded-risk convexity.

Low ERQ:

- narrow-path dependency.
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C. Liquidity Survivability

Question:

can this structure survive liquidity deformation?

D. Boundedness

Question:

is maximum damage structururally capped?

E. Re-entry Flexibility

Question:

after failure, can the system re-enter efficiently?

F. Correlation Rupture Robustness

Question:

does the exposure survive transition-law deformation?

8. New MRTP Convexity Rule

Current MRTP:

- maximize recoverability.

New MRTP:

$$\max (\text{Convexity}) \quad \text{s.t.} \quad \text{Recoverability preserved}$$

Operationally:

$$MRTP_i = \alpha R_i + \beta C_i - \gamma F_i - \delta P_i + \eta ERQ_i$$

Where:

- R_i = recoverability,
 - C_i = corridor width,
 - F_i = fragmentation,
 - P_i = ridge pressure,
 - ERQ_i = exposure recoverability quality.
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9. Kelly Integration

Current Problem

Traditional Kelly:

- expands during apparent stability,
- but stability often precedes rupture.

Thus:

- leverage expansion becomes recoverability-destructive.
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10. Recoverability-Aware Kelly

Kelly expansion permitted only when:

$$ERQ_i > \theta_{ERQ}$$

AND:

$$C_t \uparrow$$

AND:

$$F_t \downarrow$$

AND:

$$P_t \downarrow$$

Thus leverage expansion occurs only under:

- widening admissibility,
- low fragmentation,
- low ridge pressure,
- high recoverability quality.

11. Convexity Sleeve Taxonomy

Type-I Sleeves

Irreversible Convexity

Examples:

- directional leverage,
- concentrated beta,
- liquidity-fragile momentum.

Policy:

- capped,
- heavily governed,
- disabled during transition.

Type-II Sleeves

Recoverable Convexity

Examples:

- dispersion,

- bounded asymmetry,
- low-fragmentation breakout,
- optionality-preserving structures.

Policy:

- admissible,
 - scalable,
 - favored in expansion corridors.
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12. Runtime Governance Logic

LOCAL_STAR

Permit:

- recoverable convexity expansion,
- controlled Kelly increase.

Suppress:

- irreversible leverage concentration.
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TRANSITION

Prefer:

- Type-C transition sleeves,
- low-fragmentation breakouts.

Avoid:

- directional beta escalation.
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BUFFER

Freeze:

- sleeve weights,
 - leverage expansion.
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BOUNDARY / RUPTURE

Disable:

- irreversible convexity.

Maintain:

- crisis sleeves,
 - bounded optionality,
 - re-entry preservation.
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13. The Main Architectural Goal

Phase II does NOT aim to:

- maximize CAGR,
- maximize beta,
- mimic momentum.

It aims to:

materially improve admissible convex participation while preserving the crisis moat.

Meaning:

- maintain low drawdown,
- maintain recoverability,
- maintain low forced deleveraging risk,

WHILE:

- narrowing the reflation participation gap.
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14. The Fundamental MRTP Shift

Old PolyAgora:

survive rupture.

New PolyAgora:

expand only through trajectories that preserve future steering capacity.

This is the true integration of:

- MRTP,
 - recoverability geometry,
 - convexity governance,
 - admissible expansion.
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15. CTO Deliverables

Deliverable A

Exposure Recoverability Quality (ERQ) formalization.

Deliverable B

Recoverable vs irreversible sleeve classification.

Deliverable C

Recoverability-aware Kelly expansion.

Deliverable D

MRTP convexity scoring integration.

Deliverable E

Backtest:

- GFC,
- COVID,
- 2022,
- post-2020 reflation,
- momentum comparison,
- SPY correlation preservation.

Success criterion:

improve convex participation without materially degrading the drawdown moat.